Astrophysicist specialising in 21-cm cosmology and low-frequency radio astronomy, with leadership roles in HERA and EDGES, and extensive open-source software development contributions, including to the 21cmFAST simulation code. Expertise in data analysis and Bayesian inference. Experienced in large-scale collaboration, research training, and curriculum coordination.

Contact Information

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Stellenbosch University
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steven-murray
0000-0003-3059-3823

Academic References

Prof. Judd Bowman judd.bowman@asu.edu
Prof. Adrian Liu adrian.liu2@mcgill.ca
Prof. Andrei Mesinger andrei.mesinger@sns.it

Research Interests

21cm Cosmology: data analysis, seni-numerical simulations, foreground removal
Large-scale structure: halo mass function, halo model
Statistics: hierarchical Bayesian models, machine-learning methods
Software: best practices, HPC, reproducibility

Professional Experience

Senior Lecturer **2025 –**
Stellenbosch University, Stellenbosch, South Africa

MSCA (Marie Curie) Fellow **2023 – 2025**
Scuola Normale Superiore, Pisa, Italy

HERA/EDGES Postdoc **2018 – 2023**
Arizona State University, Tempe, Arizona

CAASTRO Postdoc **2015 – 2018**
Curtin University, Perth, Western Australia

APA funded PhD student **2012 – 2015**
University of Western Australia, Perth, Western Australia

Summer Internship **2011 – 2012**
ICRAR/Pawsey, Perth, Western Australia

First-year Physics Lab Demonstration Tutor **2011**
University of Western Australia, Perth, Western Australia

First-year Mathematics Tutor **2009**
University of Queensland, Brisbane, Queensland

Education

PhD, Physics **2012 – 2015**
University of Western Australia, Perth, Western Australia, Australia
- Thesis Title: Next-generation tools for next-generation surveys
- Supervisors: Prof. Chris Power, Dr. Aaron Robotham
- Area of Study: Cosmology/Structure formation

Honours, Physics **2011**
University of Western Australia, Perth, Western Australia, Australia
- Thesis Topic: Large-Scale Structure in the SDSS and GAMA surveys
- Supervisors: Prof. John Hartnett
- Area of Study: Cosmology/Structure formation

– Graduated: First Class

Bachelor of Science in Mathematics **2007 – 2009**
University of Queensland, Brisbane, Queensland, Australia
– Graduated: GPA of 6.583/7

Academic Experience

Grants **2017 –**

Stellenbosch Subcommittee B Award **2025 – 2026**
Cosmic Dawn: Will we know it when we see it?, S. G. Murray $7110

ACCESS Allocation **2024 – 2026**
Unveiling Cosmic Dawn with HERA, J. Aguirre, S.G. Murray et. al. 3M credits

Marie Curie Fellowship **2023 – 2025**
FORWARD: Forward-Models of Cosmic Dawn: connecting 21cm simulations to the real world, S. G. Murray, A. Mesinger $188,590

NASA ATP **2022 – 2024**
A new window into galaxy physics and environments during cosmic dawn through cross-correlations, S. Furlanetto, J. Mirocha, P. La Plante, D. Jacobs, S. G. Murray $473,100

NSF AAG **2022 – 2025**
Probing Cosmic Dawn with End-to-End Forward Models, S. G. Murray, J. D. Bowman, D. C. Jacobs $445,607

NSF AAG **2022 – 2024**
Collaborative Research: Exploring Reionization and the Cosmic Dawn through Cross-Correlations, P. La Plante, S. Furlanetto, S. G. Murray $257,211

XSEDE Allocation **2022 – 2024**
Unveiling Cosmic Dawn with HERA, S. G. Murray et. al. $66,009.6

XSEDE Allocation **2021 – 2022**
Unveiling Cosmic Dawn with HERA, S. G. Murray et. al. $11,960

XSEDE Startup Allocation **2020**
Unveiling Cosmic Dawn with HERA, S. G. Murray et. al.

ARC CoE **2017 – 2023**
ASTRO 3D, Lisa Kewley et al. (Murray listed as Ass. Investigator)

Collaborations **2015 –**

Network for Exploration and Space Science [CI Jack Burns] **2020 – 2023**

EDGES [CI Judd Bowman] **2019 –**

21cmFAST [CI Andrei Mesinger] **2019 –**

HERA [CI Dave DeBoer] **2018 –**

ASTRO 3D [CI Lisa Kewley] **2017 – 2018**

SKA CD/EoR SWG [CI Leon Koopmans] **2017 –**

MWA EoR Team, ICRAR/Curtin [CI Cathryn Trott] **2015 – 2018**

Memberships and Committees **2012 –**

Stellenbosch Physics Social Impact Committee [Chair] **2026 –**

HERA DE&I Committee [Chair] **2022**

HERA Ombudsperson [Ombudsperson] **2021 –**

SKA CD/EoR SWG [Member] **2017 –**

ASA [Member] **2017 –**

CAASTRO Postdoc Committee [Member] **2016 – 2018**

CAASTRO Student Committee [Chair] **2014 – 2015**

CAASTRO [Member] **2012 – 2018**

Journal Referee **2017 –**

Referee for A&C **2026 –**

Referee for PRD **2026 –**

Referee for PASA **2022 –**

Referee for A&A **2021 –**

Referee for JOSS **2019 –**

Referee for MNRAS **2017 –**

Supervision († indicates primary supervisor) **2017 –**

PhD: Fatima Saiyed **2025 –**

† Undergraduate: Naomi Carl **2023**

PhD: Daniela Breitman **2023 – 2025**

† Undergraduate: Haina Huang **2022**

† Undergraduate: Dhanush Giriyan **2021**

† Undergraduate: ASU Soft. Eng. Capstone Team **2021 – 2022**

Undergraduate: Tyler Cox **2021 – 2022**

† Undergraduate: ASU Soft. Eng. Capstone Team **2020 – 2021**

† Undergraduate: Lily Whitler **2019 – 2020**

PhD: Bella Nasirudin **2017 – 2020**

Teaching **2004 –**

Undergraduate 'CHAMP Camp': *Curriculum Coordinator* (Intensive) **2022**

Undergraduate 'Reionization Theory and 21cmFAST': *Lecturer* (CHAMP Camp) **2020 – 2023**

Undergraduate 'Advanced Python': *Lecturer* (CHAMP Camp) **2019 – 2022**

Undergraduate 'Introduction to Git and Github': *Lecturer* (CHAMP Camp) **2019 – 2022**

Undergraduate 'Physics': *Lab Demonstration; Report Grading* (UWA) **2011**

Undergraduate 'Mathematics': *Class Tutor; Assignment Grading* (UQ) **2009**

Yr 10-12 'Mathematics, Chemistry, Physics': *Private Tutor* (Private) **2006 – 2010**

Grades Pre-2 'Piano': *Private Tutor* (Private) **2004 – 2011**

Outreach **2016 –**

Outreach video for ASU Open Door, Arizona State University **2021**

Outreach stall at ASU Open Door, Arizona State University **2020**

Elementary School Presentation: "Deserts and Radio Astronomy", Eagleridge Enrichment Center **2019**

Outreach Stall at Perth Science Festival, Claremont Showgrounds **2018**

Q and A Session, Pilgrim Primary school SA via Skype **2017**

Q and A Session, Penguin District School, TAS via Skype **2017**

School Science Club Presentation: "From Plasma to Planets: How the Universe formed Structures out of Soup", Perth Modern School **2017**

CAASTRO in the Classroom Lecture: "Special Relativity", Aurora College, NSW via Skype **2017**

CAASTRO in the Classroom Lecture: "Special Relativity", NSW Schools via Skype **2016**

Industry and Inter-disciplinary Engagement **2016 –**

ASU: 21cmSense: A web-app for computing 21cm array sensitivities. *Successful proposal for and supervision of Software Engineering Capstone project team to work on web development of 21cmSense.* **2021 – 2022**

ASU: TheHaloMod: An Online Calculator for the halo model. *Successful proposal for and supervision of Software Engineering Capstone project team to work on web development of my site TheHaloMod.* **2020 – 2021**

WesCEF: Spectroscopy for soil nutrient analysis. *Consulting on data analysis of spectroscopic measurements of crops to diagnose soil nutrient issues.* **2018**

Atlassian: Atlassian ShipIT Hackathon. *Hackathon dedicated to shipping new and novel ideas in 24 hours.* **2016**

Professional Training **2013 –**

Professional Edicational Development for Academics in their Teaching Role (PREDACS), Stellenbosch University **2026**

Laboratory Safety Training, ASU **2018**

HDR Supervisor Induction, Curtin University **2017**

MWA Data Reduction Workshop, Curtin University **2016**

Code Testing for HPC (ASA Webinar Series), Virtual **2014**

Bayesian Astronomy with R, UWA **2013**

Personal Training **2015 –**

Visual Communication for Scientists **2017**

Stress Management and Resilience **2017**

The Perfect Pitch **2017**

Atlassian ShipIt Hackathon (5th place) **2016**

Conversations at the Right Wavelength **2015**

How to benefit from and contribute to Open Science **2015**

Building Strong Leaders **2015**

Creative Thinking in the Workplace **2015**

ICRAR Media Training Workshop **2015**

Awards And Scholarships

ASU
- Accepted Proposal for ASU Soft. Eng. Capstone Project (2020, for 1 year)
- Accepted Proposal for ASU Soft. Eng. Capstone Project (2021, for 1 year)

Curtin
- Most Entertaining Talk at ICRAR-CON (2017)
- Most Scientifically Challenging Talk at ICRAR-CON (2017)
- Best Overall Talk, CAASTRO Retreat (2017)

UWA
- Ernest and Evelyn Shacklock Scholarship (2012, for 3 years)
- CAASTRO Student Talk Prize (2012)
- Most Exciting Talk at ICRAR-CON (2014)

ICRAR
- ICRAR/Pawsey Summer Internship (2011, for 10 weeks)

UQ
- UQ Excellence Scholarship (2007, for 3 years)
- Dean's Commendation for High Achievement (2007, for 3 years)

Technical Skills

Proficiency with Linux (Ubuntu and Arch) operating systems.

Working knowledge of Windows and MacOS operating systems

Intimate knowledge of a variety of programming languages, in particular Python, Fortran and C, and to varying extents R, rust, HTML, CSS, Javascript and SQL.

In-depth experience with matplotlib, numpy, scipy, emcee, emacs, git, GitHub, astropy, pyyaml and h5py programs and frameworks, and to varying extents django, plotly-dash, bokeh, pandas and regex.

Software

Organizations

Most of my software development occurs within teams, which are listed here (only organizations in which I've been active are listed). Each column gives the total number of my contributions made to the organization, and then shows the relative contribution in the form of **C**ommits, **P**Rs, **I**ssues and **R**eviews.

<table>
<tr><th>Github Org</th><th>Description</th><th>Contr. [C|P|I|R]</th></tr>
<tr><td>21cmfast</td><td>Core team that develops 21cmFAST and associated pa</td><td>396 [242|15|32|107]</td></tr>
<tr><td>edges-collab</td><td>Collection of codes for working with EDGES data</td><td>358 [269|04|21|64]</td></tr>
<tr><td>HERA-Team</td><td>Collection of software used for the HERA radio tel</td><td>291 [186|15|32|58]</td></tr>
<tr><td>halomod</td><td>Halo Model calculations</td><td>174 [136|03|06|29]</td></tr>
<tr><td>rasg-affiliates</td><td>Projects affiliated with the RadioAstronomySoftwar</td><td>54 [034|00|01|19]</td></tr>
</table>

Original Codes

Here I list notable codes that I originally authored and maintain.

Key: GitHub Stars | Forks

Repo	Description	★	Forks
hmf	Python halo mass function calculator	76	35
hankel	Implementation of Ogata's (2005) method for Hankel transform	50	05
halomod	Python package for dealing with the Halo Model	36	15
powerbox	A python package for making arbitrarily structured, arbitrar	28	12
TheHaloMod-SPA	Single-page app for TheHaloMod	04	00
mrpy	A Python package for calculations with the MRP parameterisat	02	02

Collaborative Codes

Here I list notable codes that I contribute to collaboratively. They are listed in descending order of my contributions.

Key: ★ Stars | Forks | My Rank | My Contrs.

Repo	Description	★	Forks	My Contrs.	My Rank
21cmFAST	Official repository for 21cmFAST: a code for gener	76	52	1391	01/21
edges-analysis	Analysis pipeline for EDGES field data.	03	00	908	01/06
hera_sim	Simple simulation code for HERA-like redundant int	20	09	853	01/23
edges-cal	Code to calibrate EDGES data	02	00	758	01/06
hmf	Python halo mass function calculator	76	35	682	01/10
hera_cal	Library for HERA data reduction, including redunda	13	09	663	03/29
edges-io	Module for reading EDGES data and working with EDG	01	00	534	01/05
halomod	Python package for dealing with the Halo Model	36	15	471	01/08
21CMMC	Constrain 21cmFAST parameters using MCMC	08	11	444	01/08
matvis	Fast matrix-based visibility simulator	03	04	436	01/13

Presentations

Invited Talks

1. "First Results from HERA Phase II" at HIRAX Collaboration Meeting, Cape Town, South Africa (Nov 2025)
2. "First Results from HERA Phase II" at Kaba Kada, Port Douglas, Australia (Sep 2025)
3. "Data Analysis Challenges in Line Intensity Mapping: a HERA perspective" at LIM25, Annecy, France (Jun 2025)
4. "Calibration of the EDGES and HERA experiments" at Precision Calibration Workshop, Montreal, Canada (Jul 2024)
5. "Review of 21cm Cosmology Experiments" at Reionization in Relic Radiation Workshop, Paris, France (May 2024)
6. "EoR, Cosmic Dawn, and the SKA" at The Fourth National Workshop on the SKA Project, Catania, Italy (Nov 2023)
7. "EDGES3" at Global 21cm Workshop, Trieste, Italy (Sep 2023)
8. "Improved Constraints on the X-Ray Heating of the IGM from HERA Phase I" at Understanding the Epoch of Cosmic Reionization, Sesto, Italy (Mar 2023)
9. "Bayesian Insights for EDGES data" at URSI AT-AP-RASC, Gran Canaria, Canary Islands (May 2022)
10. "An Update on the Progress of EDGES" at URSI GASS, Rome, Italy (Aug 2021)

11. "Overview of new 21cmFAST and 21cmMC" at Inaugural 21cmFAST Developers Workshop, Pisa, Italy (Sep 2019)

Contributed Talks

1. "First Results from HERA Phase II" at Texas Symposium on Relativistic Astrophysics, Tempe, USA (Dec 2025)
2. "HERA Phase I and Beyond: Constraints on X-Ray Heating of the IGM" at Cosmic Dawn at High Latitudes, Stockholm, Sweden (Jun 2024)
3. "A Bayesian Calibration Framework for EDGES" at Global 21cm Workshop, Berkeley, USA (Oct 2022)
4. "An Update on the Progress of EDGES" at Global 21cm Workshop, Boulder, USA (Oct 2021)
5. "EDGES Calibration Pipeline" at Global 21cm Workshop, Cambridge, UK (Oct 2020)
6. "Current Status and Future Plans for EDGES" at Next-Generation Cosmology with Next-Generation Radio Telescopes: II, Sesto, Italy (Jan 2020)
7. "Making EDGES Bayesian" at Global 21cm Workshop, Montreal, Canada (Oct 2019)
8. "Bridging the Great Divide: Connecting Physical Foregrounds with Interferometric Instruments" at Rise and Shine, Strasbourg, France (Jun 2018)
9. "Getting the Edge on the Wedge" at ANITA Theory Workshop, Perth, Australia (Feb 2018)
10. "The Wedge and the Window" at CAASTRO Annual Retreat, Adelaide, Australia (Nov 2017) [**Prize for Best overall talk**]
11. "Between Wedge and Window: An Improved Statistical Point-Source Foreground Model for the EoR" at Peering Towards Cosmic Dawn, Dubrovnik, Croatia (Oct 2017)
12. "The Wedge and the Window" at ICRAR CON, Mandurah, Australia (Sep 2017) [**Prize for Most scientifically challenging talk and Most entertaining talk**]
13. "Between Wedge and Window: An Improved Statistical Point-Source Foreground Model for the EoR" at Fundamental Physics with the SKA, Flic-en-Flac, Mauritius (May 2017)
14. "Realistic Visibility Covariance for the EoR in the presence of... well, just about everything." at ANITA Theory Workshop, Hobart, Australia (Feb 2017)
15. "An Improved Statistical Foreground Model for the EoR" at CAASTRO Annual Retreat, Busselton, Australia (Nov 2016)
16. "A Simple Halo Mass Function Distribution" at Diving into the Dark, Cairns, Australia (Jul 2016)
17. "Eddington Bias vs. Hierarchical Bayes in the Halo Mass Function" at Statistical Challenges in 21st Century Cosmology, Chania, Greece (May 2016)
18. "Simplifying the Halo Mass Function" at ICRAR CON, Rottnest Island, Australia (Sep 2015)
19. "Dark Matters" at CAASTRO Annual Retreat, Twin Waters, Australia (Nov 2014)
20. "HALOgen: A Fast Approximate Halo Generator" at ICRAR CON, Rottnest Island, Australia (Sep 2014) [**Prize for Most Exciting Talk**]
21. "HALOgen" at nIFTy Cosmology, Madrid, Spain (Jun 2014)

22. "Tools and Statistics with Dark Matter Halos" at ANITA Theory Workshop, Sydney, Australia (Feb 2014)
23. "The Generalised 2-Point Correlation Function" at ANITA Theory Workshop, Brisbane, Australia (Feb 2013)
24. "The Generalised 2-Point Correlation Function" at CAASTRO Annual Retreat, Pinnacles, Australia (Sep 2012) [**Prize for Best Student Talk**]

Publications

To see a configurable list of all my publications, see my ADS list. Information correct as of 13 May 2026. Any arxiv e-prints displayed have been accepted. Papers in each category listed in reverse chronological order. Papers with more than 5 citations per year highlighted in orange.

At a Glance

Total Papers	79	**M-index**	2.1
Normalized Papers	10.0	**G-index**	58
Total Citations	3487	**I10-index**	56
Total Norm. Citations	409.4	**I100-index**	8
H-index	30	**Tori-index**	5.4

Key: Papers, Citations, Reads (on NASA ADS)

First/Corresponding author papers 12 793 2519

1. Abdurashidova, Zuhra, Adams, Tyrone, Aguirre, James E. et. al. (2026), *First Results from HERA Phase II*, ApJ, 998, 33 10 467
2. **SGM**, Pober, Jonathan, Kolopanis, Matthew (2024), *21cmSense v2: A modular, open-source 21 cm sensitivity calculator*, JOSS, 9, 6501 16 165
3. **SGM**, Bowman, Judd D., Sims, Peter H. et. al. (2022), *A Bayesian calibration framework for EDGES*, MNRAS, 517, 2264 27 221
4. **SGM**, Diemer, B., Chen, Z. et. al. (2021), *THEHALOMOD: An online calculator for the halo model*, A&C, 36, 100487 60 268
5. **SGM**, Greig, Bradley, Mesinger, Andrei et. al. (2020), *21cmFAST v3: A Python-integrated C code for generating 3D realizations of the cosmic 21cm signal.*, JOSS, 5, 2582 150 287
6. **SGM**, Poulin, Francis (2019), *hankel: A Python library for performing simple and accurate Hankel transformations*, JOSS, 4, 1397 19 208
7. **SGM**, Trott, C. M. (2018), *The Effect of Baseline Layouts on the Epoch of Reionization Foreground Wedge: A Semianalytical Approach*, ApJ, 869, 25 26 165
8. **SGM** (2018), *powerbox: A Python package for creating structured fields with isotropic power spectra*, JOSS, 3, 850 35 107
9. **SGM**, Robotham, A. S. G., Power, C. (2018), *An Empirical Mass Function Distribution*, ApJ, 855, 5 6 125
10. **SGM**, Trott, C. M., Jordan, C. H. (2017), *An Improved Statistical Point-source Foreground Model for the Epoch of Reionization*, ApJ, 845, 7 23 119
11. **SGM**, Power, C., Robotham, A. S. G. (2013), *HMFcalc: An online tool for calculating dark matter halo mass functions*, A&C, 3, 23 371 231
12. **SGM**, Power, C., Robotham, A. S. G. (2013), *How well do we know the halo mass function ?*, MNRAS, 434, L61 50 156

Co-authored w/ students I supervised/mentored 7 118 1649

13. Breitman, D., Mesinger, A., **SGM**, Acharya, A. (2025), *Sample variance denoising in cylindrical 21 cm power spectra*, Astronomy and Astrophysics, 703, A130 0 190
14. Breitman, Daniela, Mesinger, Andrei, **SGM** et. al. (2024), *21CMEMU: an emulator of 21CMFAST summary observables*, MNRAS, 527, 9833 38 420
15. Vydula, Akshatha K., Bowman, Judd D., Lewis, David et. al. (2024), *Low-frequency Radio Recombination Lines Away from the Inner Galactic Plane*, The Astronomical Journal, 167, 2 3 250
16. Cox, Tyler A., Jacobs, Daniel C., **SGM** (2022), *Estimating the feasibility of 21cm-Ly synergies using the hydrogen Epoch of Reionization array*, MNRAS, 512, 792 10 212
17. Lanman, Adam E., **SGM**, Jacobs, Daniel C. (2022), *Validation Solutions to the Full-sky Radio Interferometry Measurement Equation for Diffuse Emission*, The Astrophysical Journal Supplement Series, 259, 22 9 166
18. Mahesh, Nivedita, Bowman, Judd D., Mozdzen, Thomas J. et. al. (2021), *Validation of the EDGES Low-band Antenna Beam Model*, The Astronomical Journal, 162, 38 40 280
19. Nasirudin, A., **SGM**, Trott, C. M. et. al. (2020), *The Impact of Realistic Foreground and Instrument Models on 21 cm Epoch of Reionization Experiments*, ApJ, 893, 118 18 131

Significant contribution to analysis 27 688 5538

20. Sims, Peter H., Bowman, Judd D., **SGM** et. al. (2025), *A Bayesian approach to modelling spectrometer data chromaticity corrected using beam factors II. Model priors and posterior odds*, MNRAS, 544, 2340 8 329
21. Davies, James E., Mesinger, Andrei, **SGM** (2025), *Efficient simulation of discrete galaxy populations and associated radiation fields over the first billion years*, Astronomy and Astrophysics, 701, A236 15 210
22. Sims, Peter H., Bowman, Judd D., **SGM** et. al. (2025), *A general Bayesian model-validation framework based on null-test evidence ratios, with an example application to global 21-cm cosmology*, MNRAS, 541, 2262 7 205
23. Cang, Junsong, Mesinger, Andrei, **SGM** et. al. (2025), *The EDGES measurement disfavors an excess radio background during the cosmic dawn*, Astronomy and Astrophysics, 698, A152 13 234
24. Keating, Garrett, Hazelton, Bryna, Kolopanis, Matthew et. al. (2025), *pyuvdata v3: an interface for astronomical interferometric data sets in Python*, JOSS, 10, 7482 9 0
25. Cox, Tyler A., **SGM**, Parsons, Aaron R. et. al. (2025), *fftvis: a non-uniform Fast Fourier Transform based interferometric visibility simulator*, RAS Techniques and Instruments, 4, rzaf056 2 284
26. Kittiwisit, Piyanat, **SGM**, Garsden, Hugh et. al. (2025), *matvis: a matrix-based visibility simulator for fast forward modelling of many-element 21 cm arrays*, RAS Techniques and Instruments, 4, rzaf001 13 207
27. Sims, Peter H., Bowman, Judd D., Mahesh, Nivedita et. al. (2023), *A Bayesian approach to modelling spectrometer data chromaticity corrected using beam factors - I. Mathematical formalism*, MNRAS, 521, 3273 22 259
28. Gorce, Adélie, Ganjam, Samskruthi, Liu, Adrian et. al. (2023), *Impact of instrument and data characteristics in the interferometric reconstruction of the 21 cm power spectrum*, MNRAS, 520, 375 14 346

29. Kittiwisit, Piyanat, Bowman, Judd D., **SGM** et. al. (2022), *Measurements of one-point statistics in 21-cm intensity maps via foreground avoidance strategy*, MNRAS, 517, 2138 6 291
30. Greig, Bradley, Wyithe, J. Stuart B., **SGM** et. al. (2022), *Generating extremely large-volume reionization simulations*, MNRAS, 516, 5588 15 179
31. Trott, Cathryn M., Mondal, Rajesh, Mellema, Garrelt et. al. (2022), *Multi-frequency angular power spectrum of the 21 cm signal from the Epoch of Reionisation using the Murchison Widefield Array*, Astronomy and Astrophysics, 666, A106 10 162
32. Nasirudin, Ainulnabilah, Prelogovic, David, **SGM** et. al. (2022), *Characterizing beam errors for radio interferometric observations of reionization*, MNRAS, 514, 4655 4 173
33. Mondal, Rajesh, Mellema, Garrelt, **SGM**, Greig, Bradley (2022), *The multifrequency angular power spectrum in parameter studies of the cosmic 21-cm signal*, MNRAS, 514, L31 17 152
34. Muñoz, Julian B., Qin, Yuxiang, Mesinger, Andrei et. al. (2022), *The impact of the first galaxies on cosmic dawn and reionization*, MNRAS, 511, 3657 132 298
35. Aguirre, James E., **SGM**, Pascua, Robert et. al. (2022), *Validation of the HERA Phase I Epoch of Reionization 21 cm Power Spectrum Software Pipeline*, ApJ, 924, 85 30 242
36. Prelogović, David, Mesinger, Andrei, **SGM** et. al. (2022), *Machine learning astrophysics from 21 cm lightcones: impact of network architectures and signal contamination*, MNRAS, 509, 3852 44 265
37. Gehlot, Bharat K., Jacobs, Daniel C., Bowman, Judd D. et. al. (2021), *Effects of model incompleteness on the drift-scan calibration of radio telescopes*, MNRAS, 506, 4578 8 218
38. Chen, Zhaoting, Wolz, Laura, Spinelli, Marta, **SGM** (2021), *Extracting H I astrophysics from interferometric intensity mapping*, MNRAS, 502, 5259 22 171
39. Trott, Cathryn M., Watkinson, Catherine A., Jordan, Christopher H. et. al. (2019), *Gridded and direct Epoch of Reionisation bispectrum estimates using the Murchison Widefield Array*, PASA, 36, e023 30 200
40. Trott, Cathryn M., Fu, Shih Ching, **SGM** et. al. (2019), *Robust statistics towards detection of the 21 cm signal from the Epoch of Reionization*, MNRAS, 486, 5766 8 148
41. Wolz, L., **SGM**, Blake, C., Wyithe, J. S. (2019), *Intensity mapping cross-correlations II: HI halo models including shot noise*, MNRAS, 484, 1007 39 190
42. Meyers, B. W., Tremblay, S. E., Bhat, N. D. R. et. al. (2018), *Hunting for Radio Emission from the Intermittent Pulsar J1107-5907 at Low Frequencies*, ApJ, 869, 134 21 185
43. Trott, Cathryn M., Jordan, C. H., **SGM** et. al. (2018), *Assessment of Ionospheric Activity Tolerances for Epoch of Reionization Science with the Murchison Widefield Array*, ApJ, 867, 15 30 0
44. Obreschkow, D., **SGM**, Robotham, A. S. G., Westmeier, T. (2018), *Eddington's demon: inferring galaxy mass functions and other distributions from uncertain data*, MNRAS, 474, 5500 33 220
45. Jordan, C. H., **SGM**, Trott, C. M. et. al. (2017), *Characterization of the ionosphere above the Murchison Radio Observatory using the Murchison Widefield Array*, MNRAS, 471, 3974 74 157
46. Avila, Santiago, **SGM**, Knebe, Alexander et. al. (2015), *HALOGEN: a tool for fast generation of mock halo catalogues*, MNRAS, 450, 1856 62 213

Collaboration papers (contr. to analysis and/or writing) 33 1888 8375

47. Cappallo, Rigel C., Rogers, Alan E. E., Lonsdale, Colin J. et. al. (2025), *EDGES-3: Instrument Design and Commissioning*, Publications of the Astronomical Society of the Pacific, 137, 125002 0 389
48. Kim, Honggeun, Hewitt, Jacqueline N., Kern, Nicholas S. et. al. (2025), *Exploring One-point Statistics in HERA Phase I Data: Effects of Foregrounds and Systematics on Measuring One-point Statistics*, ApJ, 993, 189 2 224
49. Bonaldi, A., Hartley, P., Braun, R. et. al. (2025), *Square Kilometre Array Science Data Challenge 3a: foreground removal for an EoR experiment*, MNRAS, 543, 1092 13 383
50. Rath, E., Pascua, R., Josaitis, A. T. et. al. (2025), *Investigating mutual coupling in the hydrogen epoch of reionization array and mitigating its effects on the 21-cm power spectrum*, MNRAS, 541, 1125 26 235
51. Chen, Kai-Feng, Wilensky, Michael J., Liu, Adrian et. al. (2025), *Impacts and Statistical Mitigation of Missing Data on the 21 cm Power Spectrum: A Case Study with the Hydrogen Epoch of Reionization Array*, ApJ, 979, 191 12 217
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